CS19541 - COMPUTER NETWORKS LAB

EXPERIMENT NO: 09

AIM

Implementation of SUBNETTING in CISCO PACKET TRACER simulator.

CREATING A NETWORK TOPOLOGY

Classless IP subnetting allows efficient utilization of IP address space by defining custom subnet masks. To construct the topology in Cisco Packet Tracer:

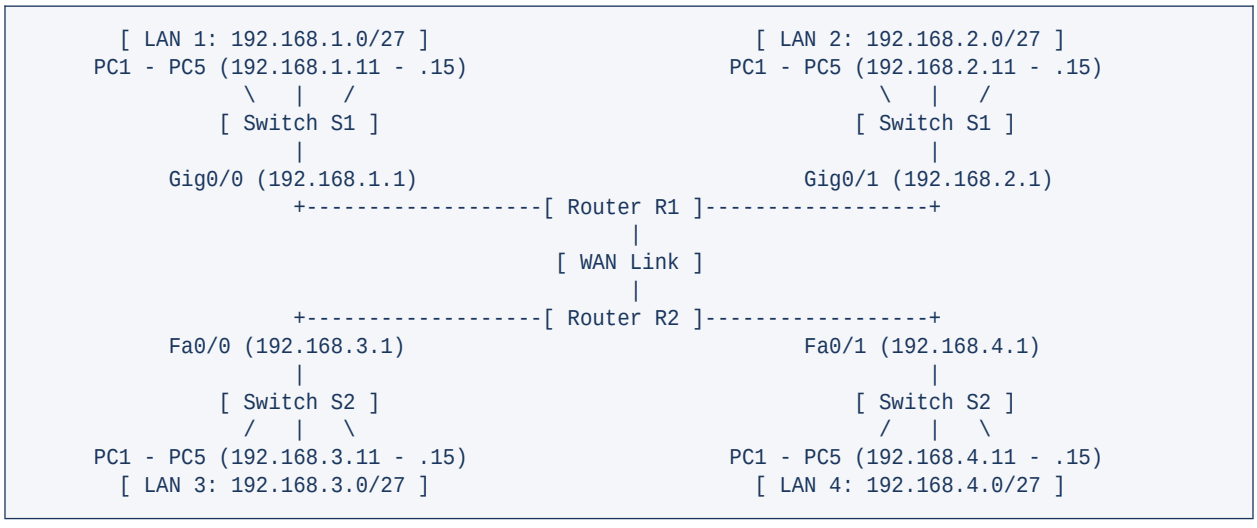
- Open Cisco Packet Tracer and create a blank workspace by selecting New > Network > Generic.
- Add Devices: Drag and drop 2 Routers (Router R1, Router R2), 2 Switches (Switch S1, Switch S2), and 20 PCs (5 PCs per LAN subnet) onto the canvas.
- Interconnect Devices: Connect PCs to Switch access ports using Copper Straight-Through cables. Connect Switches to designated Router interfaces using Copper Straight-Through cables, and establish a WAN link between Router R1 and Router R2.

SUBNETTING & TOPOLOGY DIAGRAM

To subnet the network address 192.168.1.0/24 to accommodate at least 5 host addresses per network along with router/switch interfaces, a /27 subnet mask (255.255.255.224) is used:

- **Base Subnet Mask:** 255.255.255.224 (/27)
- **Total Subnets Generated:** 8 subnets
- **Usable Host Addresses per Subnet:** 30 host addresses (2^5 - 2)

Network Topology Architecture:



DEVICE IP ADDRESSING SCHEME

Device	Interface / Port	Assigned IP Address / Subnet	Default Gateway
Router R1	GigabitEthernet0/0	192.168.1.1 /27	N/A
Router R1	GigabitEthernet0/1	192.168.2.1 /27	N/A
Switch S1 (LAN 1)	FastEthernet0/1 (PC1 - PC5)	192.168.1.11 - 192.168.1.15	192.168.1.1
Switch S1 (LAN 2)	FastEthernet0/2 (PC1 - PC5)	192.168.2.11 - 192.168.2.15	192.168.2.1
Router R2	FastEthernet0/0	192.168.3.1 /27	N/A
Router R2	FastEthernet0/1	192.168.4.1 /27	N/A
Switch S2 (LAN 3)	FastEthernet0/1 (PC1 - PC5)	192.168.3.11 - 192.168.3.15	192.168.3.1
Switch S2 (LAN 4)	FastEthernet0/2 (PC1 - PC5)	192.168.4.11 - 192.168.4.15	192.168.4.1

TESTING THE NETWORK

Network connectivity was verified using Cisco Packet Tracer simulation PDU transmission across devices and subnets. The test verification table is recorded below:

Fire / Status	Last Status	Source	Destination	Type	Time (sec)	Periodic	Num
Normal	Successful	PC4(2)	Router2	ICMP	0.000	N	12
Normal	Successful	PC4(2)	PC2(1)(1)	ICMP	0.000	N	13
Normal	Successful	PC0	Router0	ICMP	0.000	N	14

RESULT

The network topology was successfully designed, configured, and classless IP subnetting (/27) was implemented using the Cisco Packet Tracer simulator. Inter-subnet and intra-subnet communication between all hosts, switches, and routers was verified and tested successfully using ICMP ping packets.